

Average speed

Average speed is a ratio: the distance covered divided by the time taken. It teaches average speed and assumes you can already read a ratio.

Write Δs for the distance covered and Δt for the time it took. Then the average speed is

$$\bar{v} = \frac{\Delta s}{\Delta t}.$$

A journey of $\Delta s = 150 \text{ km}$ taking $\Delta t = 2 \text{ h}$ has an average speed of 75 km/h .

What the average leaves out

The average says nothing about how the speed varied. The same 75 km/h average describes a steady cruise and a journey that alternated between a standstill and 150 km/h .

Because it is a ratio, scaling both quantities together leaves it unchanged: 300 km in 4 h is still 75 km/h . That is the property carried over from ratios, not a separate fact about motion.

Units follow the division

The unit of an average speed is the unit of distance divided by the unit of time. Dividing kilometres by hours gives km/h ; dividing metres by seconds gives m/s . A speed quoted without its unit is not a speed.

Speed from a distance-time graph

A distance-time graph plots distance on the vertical axis against time on the horizontal one. Reading average speed off it is the same division as before, done geometrically.

Between two instants t_1 and t_2 , the average speed is the slope of the straight line joining the two points:

$$\bar{v} = \frac{s(t_2) - s(t_1)}{t_2 - t_1}.$$

A steeper line means a larger speed. A horizontal line means the distance is not changing, so the object is at rest.

From the average to the instant

Shrinking the interval around a single instant turns the chord into the tangent at that point, and the average speed into the instantaneous speed:

$$\begin{aligned} v(t) &= \lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t} \\ &= \frac{ds}{dt}. \end{aligned}$$

The average over a whole journey is a single number. The tangent gives one number per instant, which is why a graph says more than an average does.

Reading it honestly

A distance-time graph records distance travelled, not position, so it never decreases. A curve bending upwards means the speed is increasing; a curve flattening out means it is decreasing.

Balanced forces

The everyday intuition is that keeping something moving requires a continuous push. On a level surface without friction it does not. What a push changes is the velocity, not the motion itself.

Forces add as vectors, so the net force on an object is

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i .$$

When $\vec{F}_{\text{net}} = \vec{0}$ the forces are balanced and the velocity does not change: an object at rest stays at rest, and one already moving keeps the same speed in the same direction.

Why the intuition feels right

A box pushed across a floor stops because friction is there, not because motion needs a supply of force. The push is balancing friction rather than sustaining the motion. Remove the friction and the same push would keep speeding the box up.

What a net force does

An unbalanced force changes velocity in proportion to itself and in inverse proportion to the mass:

$$\vec{a} = \frac{\vec{F}_{\text{net}}}{m}.$$

Acceleration is a change of velocity, which includes a change of direction. Something moving at a constant speed around a circle is accelerating, because its direction keeps changing.